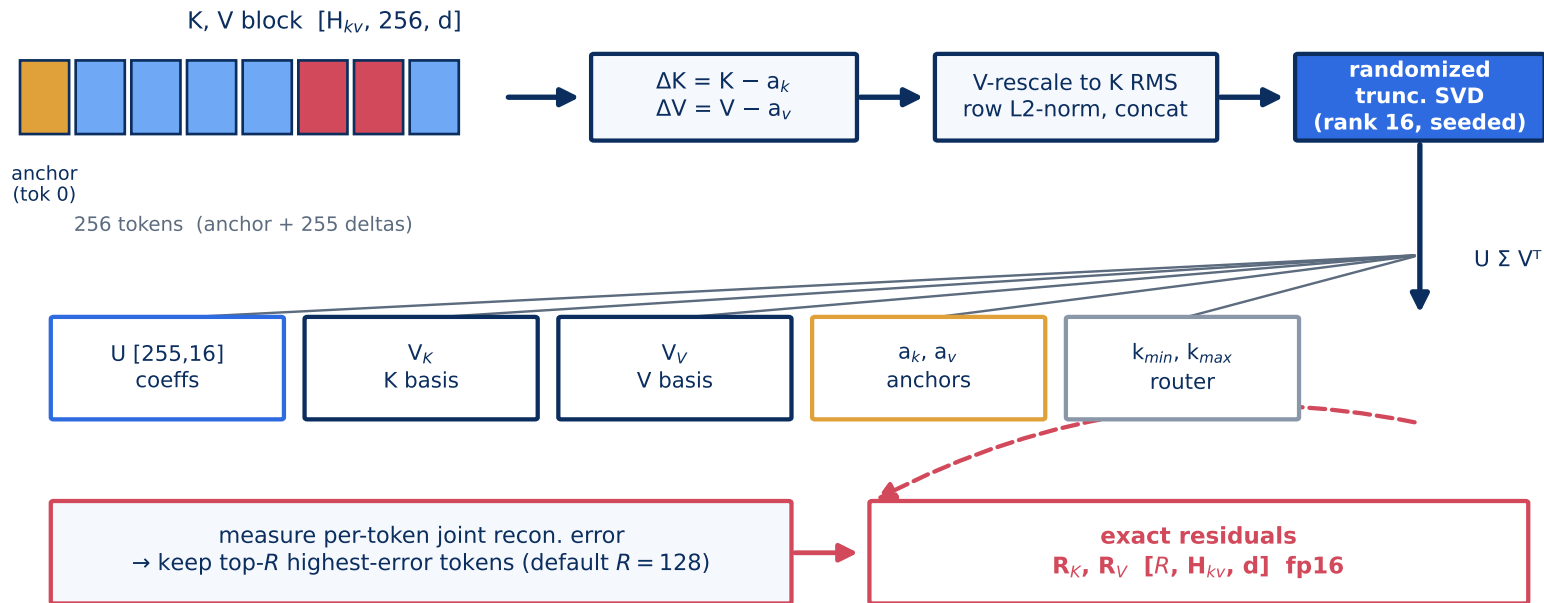


Differential KV Compression — one 256-token block

Anchor + low-rank joint SVD delta (shared structure) + top-error exact residuals (the outliers SVD discards)



stored block ≈ 154 KiB (26 KiB low-rank + 128 KiB residuals, $R = 128$) vs 256 KiB dense $\rightarrow 1.66\times$ smaller

$R = 64$ memory preset $\rightarrow 90$ KiB, $2.85\times$. Low-rank recon (implicit): $\hat{K}_i \approx a_k + s(U_i V_K)$; residuals attended EXACTLY at decode.